

Inventions

The work of inventors is all around you. Not just your phone and games console – the chair you are sitting on, the car outside, even the light bulb above your head was invented by somebody. Some early inventions, like the wheel, will be used for ever. Others, such as the spear, have been replaced by newer, more effective models.

▶ 1,760,000 BCE



FLINT HAND AXE

HAND AXE

Flint is a special kind of rock because it breaks into sharp pieces. Stone-Age people discovered that its hard, sharp edges made it very useful as a tool. Shaped into an axe, it could be used for cutting meat, scraping skins (to make clothes), chopping wood, and as a weapon.

▶ 35,000 BCE

SPEAR

The problem with hand-held weapons was that hunters had to stand very close to their prey, which was dangerous. The invention of the spear solved this problem. The hunter could stand back some distance, take aim, and throw the weapon. Early spears had flint heads. Later ones used metal heads, shaped into long, thin blades.



SHORT SPEARS

▶ 17,500 BCE

ANCIENT EGYPTIAN JAR



POTTERY

Chinese inventors realized they could dig clay from the ground, shape it into pots, and harden them in hot ashes. The pots were watertight so they could be used to carry or heat up water and food.

◀ 1876

▶ 1862



ROTARY PHONE

TELEPHONE

Early in the 19th century people found they could send signals through wires, but it was not until the invention of the telephone by Alexander Graham Bell in 1876 that voices could be sent along wires at long distance. This invention revolutionized the ways in which we communicate.

PLASTIC

British inventor Alexander Parkes was trying to create a synthetic material that could be easily shaped when hot, but would be hard when cold. In 1862 he exhibited Parkesine, the world's first type of plastic.



MODERN PLASTIC BOTTLES

◀ 1834

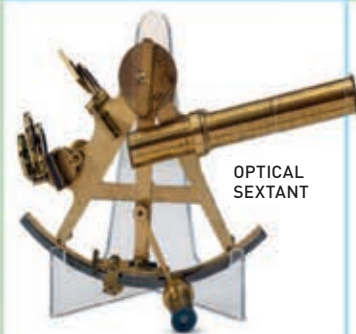
REFRIGERATOR

Until 1834 people kept food cool in insulated boxes filled with ice, which was delivered to their door. Then Jacob Perkins of Philadelphia, USA, invented a water-freezing machine that led to the first domestic fridge.



1950s REFRIGERATOR

◀ 1759



OPTICAL SEXTANT

SEXTANT

As explorers continued their long journeys across oceans, there was a need for accurate instruments for navigation. In 1759 British instrument maker John Bird perfected the sextant, which is still kept on ships today as a back-up device in case GPS (satnav) navigation fails.

◀ 1712



▶ 1878

▶ 1886



EDISON'S LAMP

LIGHT BULB

Scientists across the world experimented with lamps and light in the 19th century, but it was Thomas Edison in the USA, who created a light bulb that could last for more than 1,200 hours. Light bulbs have since been redesigned to use less energy.



1900 BENZ IDEAL

CAR

Karl Benz of Germany built the first stationary petrol engine in 1879, and decided to work out how to use this in a "horseless carriage". By 1885 he had invented a two-seater vehicle with a compact, single-cylinder engine. The patent for this car, filed in 1886, is seen as the "birth certificate" of the motor car.

▶ 1895



CABINET WIRELESS RADIO, 1932

RADIO COMMUNICATION

In 1895 Italian inventor Guglielmo Marconi managed to send Morse code signals using radio waves instead of wires. The instrument he used became known as the radio.

▶ 1903

AEROPLANE

Orville Wright from the USA first took to the skies with an aeroplane powered by a small petrol engine in North Carolina in 1903. He flew for 12 seconds over a distance of 37 m (120 ft). He and his brother Wilbur had spent five years in their workshop in Ohio designing machines that were strong, light, and had enough balance and control to fly.



MODEL OF THE WRIGHTS' 1903 FLYER

▶ 1923



TV SET FROM THE 1950s

TELEVISION

John Logie Baird, from Scotland, was the first person to transmit a TV picture in 1923. In 1927 American Philip Farnsworth created the first form of electronic television.